Advanced Macroeconomics

I. Foundations of Dynamic Macroeconomic Modeling

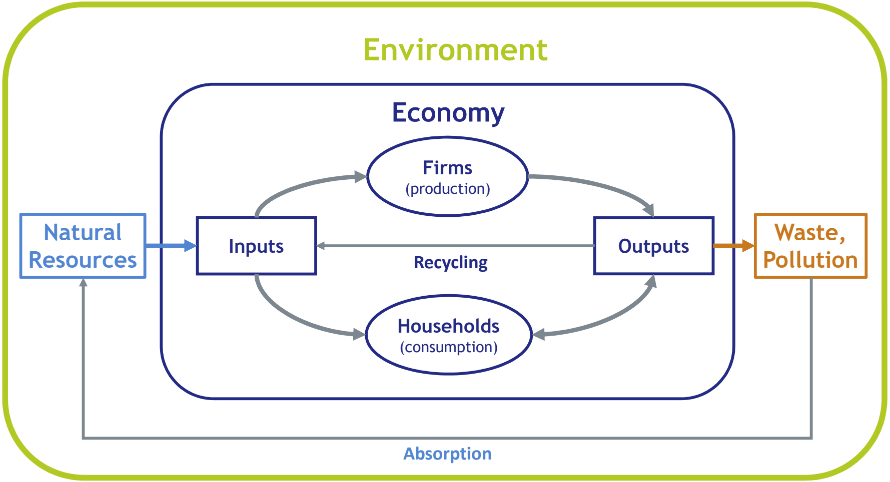
II. Long-run Economic Growth

III. Short-run Fluctuations

IV. Applications

11. Fiscal Policy and Economic Growth
12. Fiscal Policy and Business Cycles
13. Model Estimation and Macroeconomic Forecasting
14. Integrated Assessment Modeling

Environment and Economy



14. Integrated Assessment Modeling

Drivers of CO_2-Emissions
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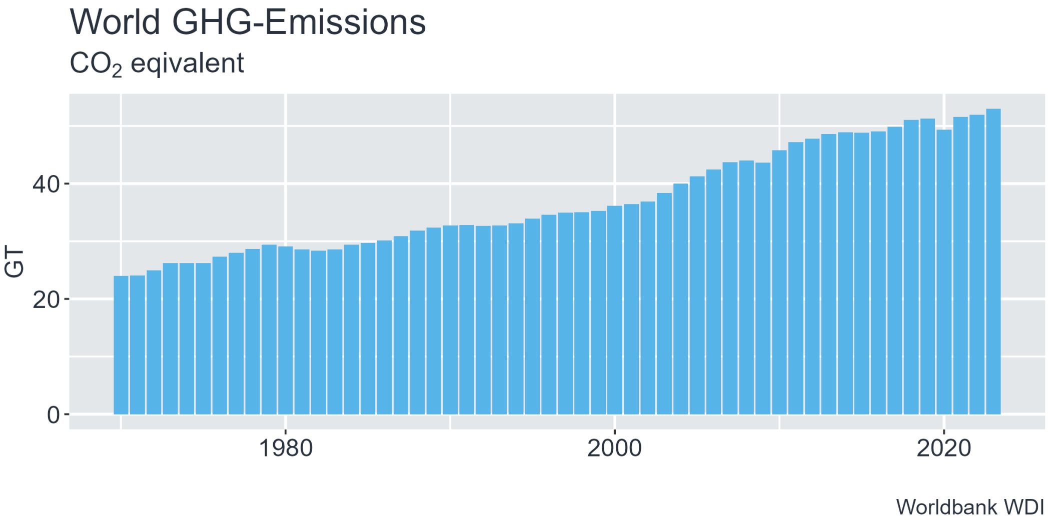
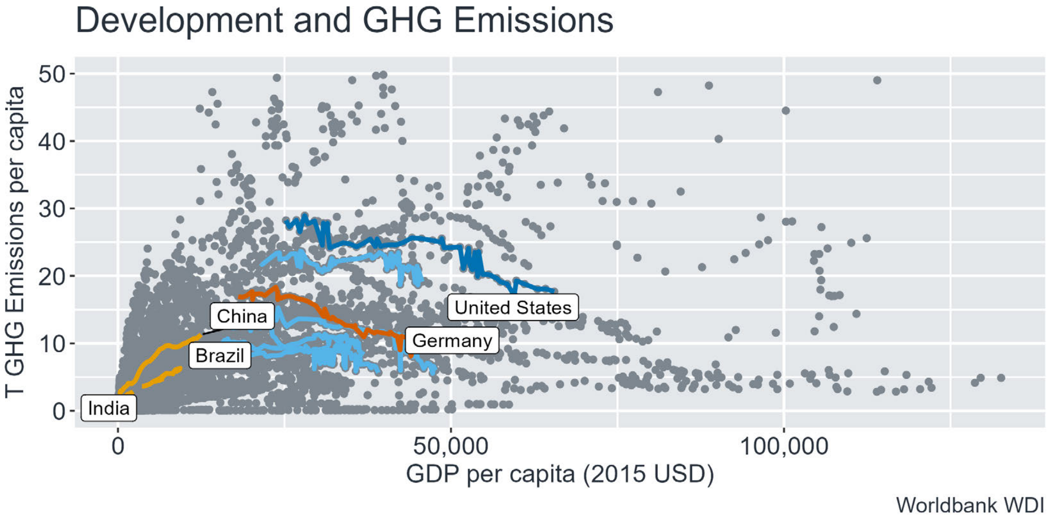
CO_2-Emissions and Directed Technical Change
Summary and Literature

CO_2-Emissions

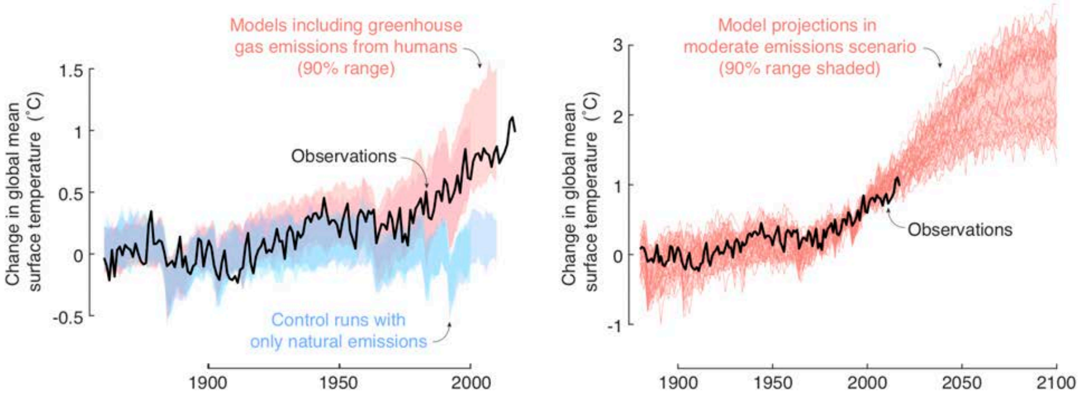


GDP per Capita and CO_2-Emissions



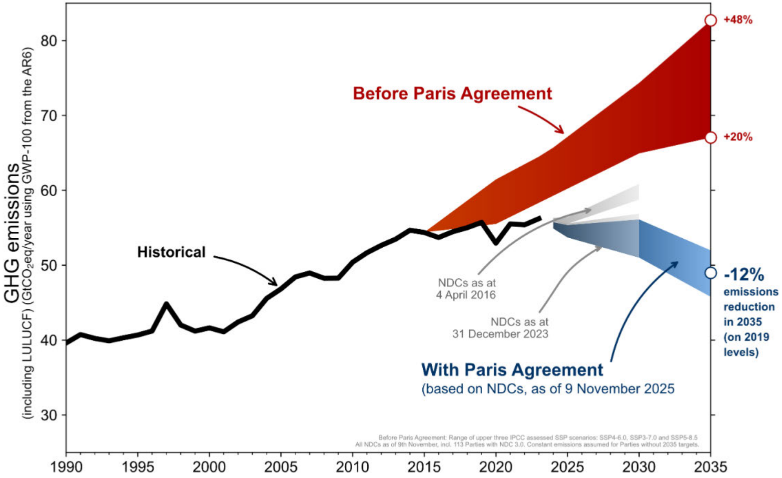
Global Warming

(Hsiang und Kopp 2018)



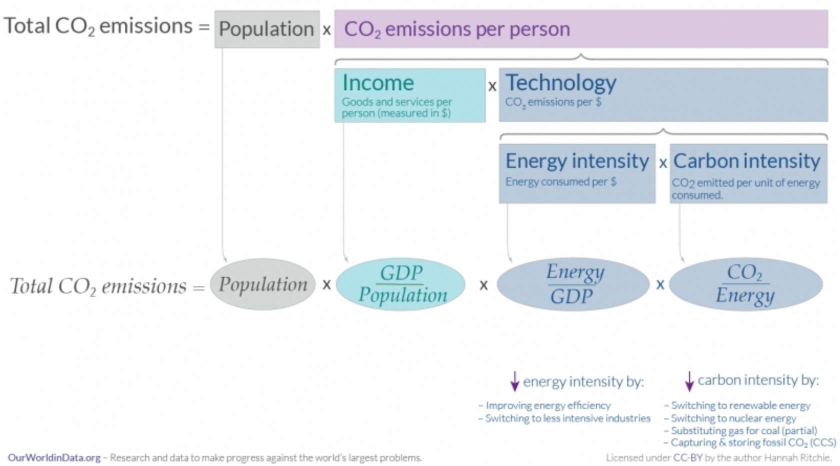
Global Emission Reduction Targets

(United Nations)



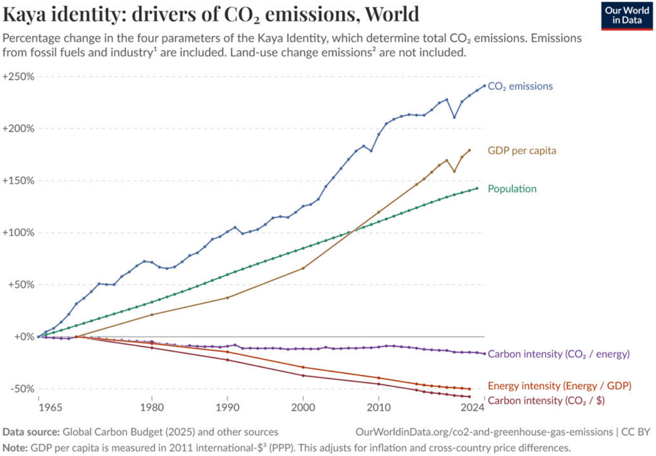
Kaya-Identity: What Drives CO_2-Emissions?

https://ourworldindata.org/emissions-drivers



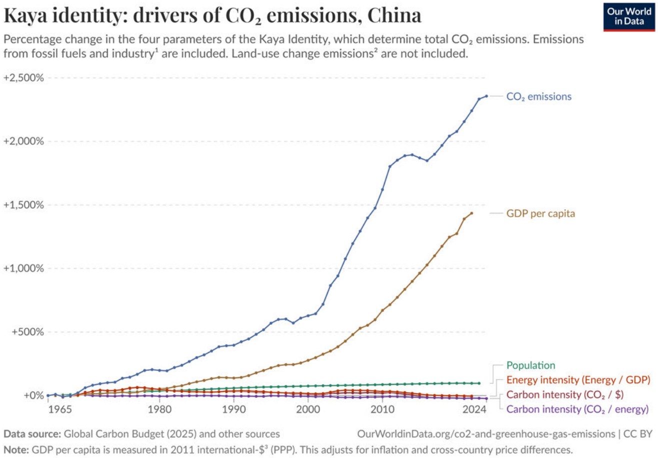
Percentage Change in the Parameters of the Kaya Identity, World

(https://ourworldindata.org/emissions-drivers)



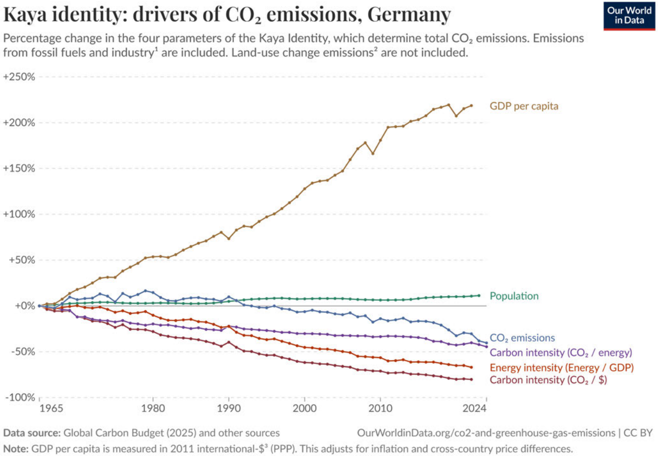
Percentage Change in the Parameters of the Kaya Identity, China

(https://ourworldindata.org/emissions-drivers)



Percentage Change in the Parameters of the Kaya Identity, Germany

(https://ourworldindata.org/emissions-drivers)



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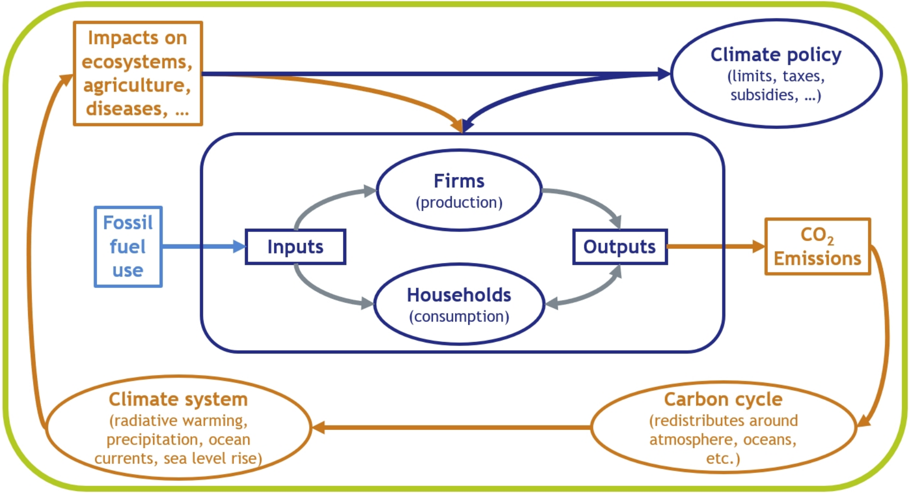
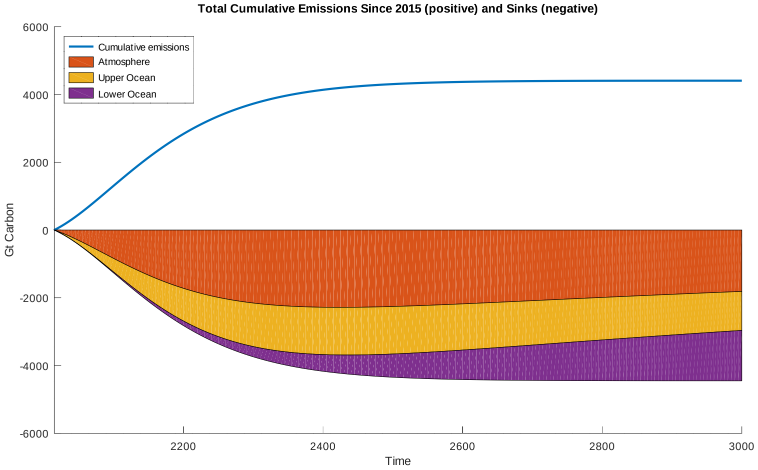
Integrated Assessment Models

(Nordhaus 2013)



Absorption of CO_2-Emissions

(Nordhaus 2013, own calculations)



DICE Model: Carbon Cycle

- One period t consists of $\tau = 5$ years
- Total annual CO_2 emissions ($ETOT$) consist of industrial and land-use emissions ($ETREE$)

$$ETOT_t = EY_t + ETREE_t$$

- One ton of CO_2 contains 1/3.666 tons of carbon
- The carbon cycle is based upon a three-reservoir model calibrated to existing carbon-cycle models and historical data

$$\begin{aligned}
MAT_t &= ETOT_t \times \tau/3.666 + b_{11}MAT_{t-1} + b_{21}MUP_{t-1} \\
MUP_t &= b_{12}MAT_{t-1} + b_{22}MUP_{t-1} + b_{32}MLO_{t-1} \\
MLO_t &= b_{23}MUP_{t-1} + b_{33}MLO_{t-1}
\end{aligned}$$

DICE Model: Warming

- Accumulations of carbon in the atmosphere lead to warming through increases in radiative forcing

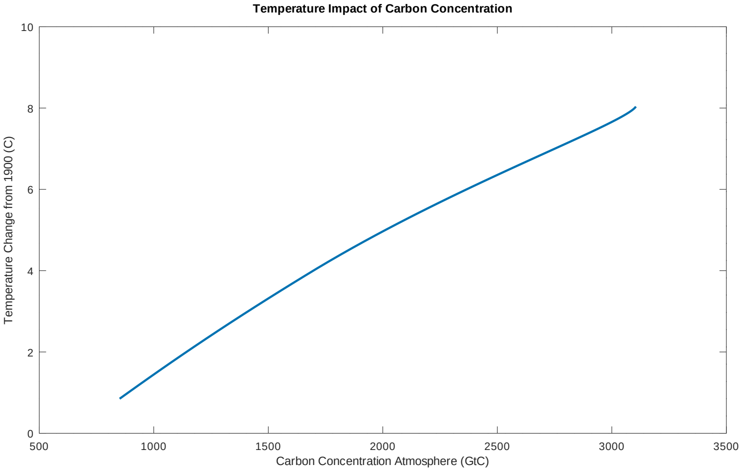
$$FORC_t = \eta \log_2 \frac{MAT_t}{MAT_{1750}} + FORCOTH_t$$

- *FORC* is the change in total radiative forcings of greenhouse gases since 1750 from anthropogenic sources such as CO_2; *FORCOTH* is exogenous forcings
- Temperature impact

$$\begin{aligned} TATM_t &= TATM_{t-1} + c_1(FORC_t - c_2 TATM_{t-1}) - c_3(TATM_{t-1} - TOCEAN_{t-1}) \\ TOCEAN_t &= TOCEAN_{t-1} + c_4(TATM_{t-1} - TOCEAN_{t-1}) \end{aligned}$$

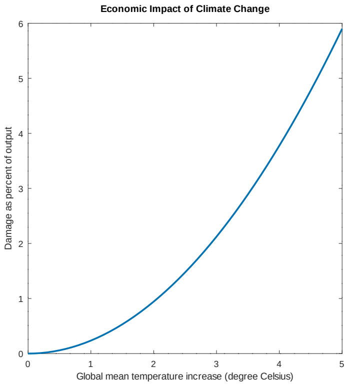
Carbon Concentration and Warming

(Nordhaus 2013, own calculations)



Economic Damages of Global Warming

(Acevedo et al. 2018, Hassler und Krusell 2018, own calculations)



- Channels
  - Agricultural production
  - Coast damages
  - Health problems
  - Frequency of extreme weather events
- Damage function

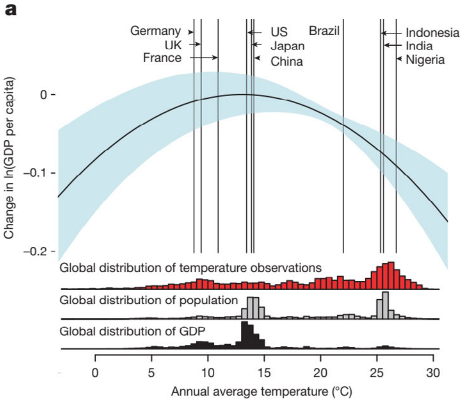
$$DAMFRAC_t = a_1 \, TATM_t + a_2 \, TATM_t^{a_3}$$

- Net output

$$YNET_t = (1 - DAMFRAC_t) \times YGROSS_t$$

Economic Effects of Climate Change Depend on Initial Temperature

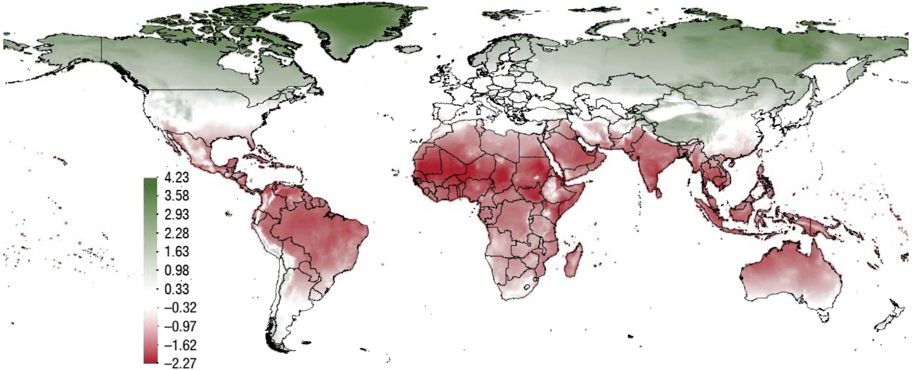
(Burke, Hsiang, Miguel 2015)



Global North is Winning – Global South is Losing

(Acevedo et al. 2018)

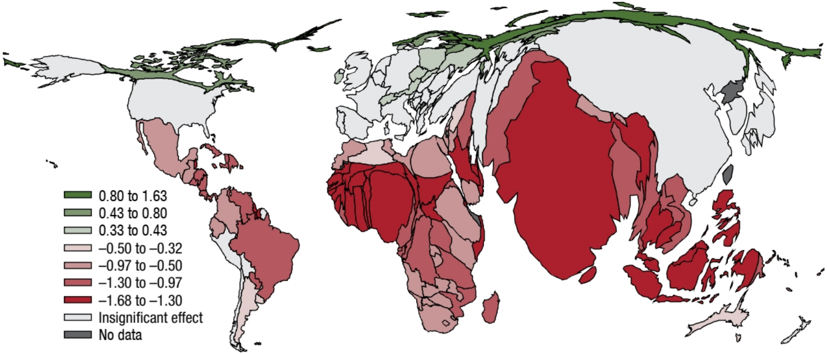
1. Effect of a 1°C Increase in Temperature on Real per Capita Output at the Grid Level



Adjusted for Population Negative Effects Larger than Positive Effects

(Acevedo et al. 2018)

2. Effect of a 1°C Increase in Temperature on Real per Capita Output at the Country Level, with Countries Rescaled in Proportion to Their Population



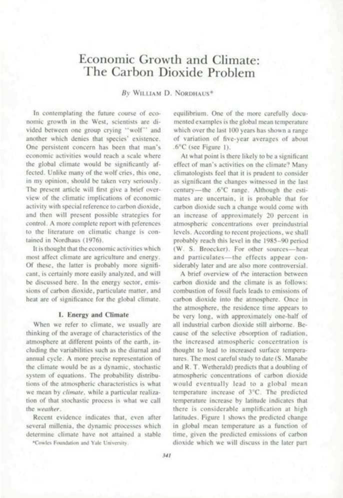
Climate Policy and Economic Growth: William Nordhaus

William D. NORDHAUS (geb. 1941)

2018 The Sveriges Riksbank Prize in Economic Sciences in Memory

of Alfred Nobel

https://www.nobelprize.org/prizes/economic-sciences/2018/nordhaus/facts/



Economic Growth and Climate: The Carbon Dioxide Problem

By WILLIAM D. NORDHAUS*

In contemplating the future course of economic growth in the West, scientists are divided between one group crying "wolf" and another which denies that species' existence. One persistent concern has been that man's economic activities would reach a scale where the global climate would be significantly affected. Unlike many of the wolf cries, this one, in my opinion, should be taken very seriously. The present article will first give a brief overview of the climatic implications of economic activity with special reference to carbon dioxide, and then will present possible strategies for control. A more complete report with references to the literature on climatic change is contained in Nordhaus (1976).

It is thought that the economic activities which most affect climate are agriculture and energy. Of these, the latter is probably more significant, is certainly more easily analyzed, and will be discussed here. In the energy sector, emissions of carbon dioxide, particulate matter, and heat are of significance for the global climate.

I. Energy and Climate

When we refer to climate, we usually are thinking of the average of characteristics of the atmosphere at different points of the earth, including the variabilities such as the diurnal and annual cycle. A more precise representation of the climate would be as a dynamic, stochastic system of equations. The probability distributions of the atmospheric characteristics is what we mean by *climate*, while a particular realization of that stochastic process is what we call the *weather*.

Recent evidence indicates that, even after several millenia, the dynamic processes which determine climate have not attained a stable equilibrium. One of the more carefully documented examples is the global mean temperature which over the last 100 years has shown a range of variation of five-year averages of about .6°C (see Figure 1).

At what point is there likely to be a significant effect of man's activities on the climate? Many climatologists feel that it is prudent to consider as significant the changes witnessed in the last century—the .6°C range. Although the estimates are uncertain, it is probable that for carbon dioxide such a change would come with an increase of approximately 20 percent in atmospheric concentrations over preindustrial levels. According to recent projections, we shall probably reach this level in the 1985–90 period (W. S. Broecker). For other sources—heat and particulates—the effects appear considerably later and are also more controversial.

A brief overview of the interaction between carbon dioxide and the climate is as follows: combustion of fossil fuels leads to emissions of carbon dioxide into the atmosphere. Once in the atmosphere, the residence time appears to be very long, with approximately one-half of all industrial carbon dioxide still airborne. Because of the selective absorption of radiation, the increased atmospheric concentration is thought to lead to increased surface temperatures. The most careful study to date (S. Manabe and R. T. Wetherald) predicts that a doubling of atmospheric concentrations of carbon dioxide would eventually lead to a global mean temperature increase of 3°C. The predicted temperature increase by latitude indicates that there is considerable amplification at high latitudes. Figure 1 shows the predicted change in global mean temperature as a function of time, given the predicted emissions of carbon dioxide which we will discuss in the later part

*Cowles Foundation and Yale University.

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(American Economic Review 67(1), 1977)

Dynamic Integrated Model of Climate and the Economy (DICE)

(William Nordhaus)

- Solow Growth Model: L, A exogenous; $YGROSS$, K, endogenous
- CO_2-Emissions endogenous
- Temperature endogenous depending on accumulated CO_2 in the atmosphere (carbon cycle)
- Economic damage depending on temperature
- Emission control is a political decision, causing mitigation cost
- Social welfare function

$$W_0 = \sum_{t=0}^{T} \frac{u(c_t)L_t}{(1+\rho)^t}$$

Emission Control

- Emission efficiency

$$X_t = (1 + \gamma_x) X_{t-1}, \qquad \gamma_x < 0$$

- Mitigation (*MIU* = emission reduction rate)

$$EY_t = (1 - MIU_t) \times X_t \times YGROSS_t$$

- Abatement

$$ABATECOST_t = \theta_{1,t} MIU_t^{\theta_2} \times YGROSS_t, \qquad \theta_{1,t} = PBACK_t \times X_t / \theta_2 / 1000$$

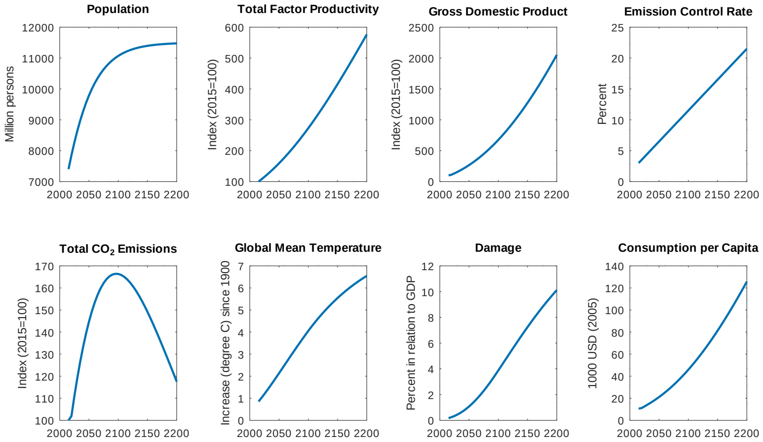
where *PBACK* denotes cost of backstop 2010 USD per ton CO_2. Backstop technology is a technology that can replace all fossil fuels.

- Net output after damages and abatement

$$Y_t = YNET_t - ABATECOST_t = C_t + I_t$$

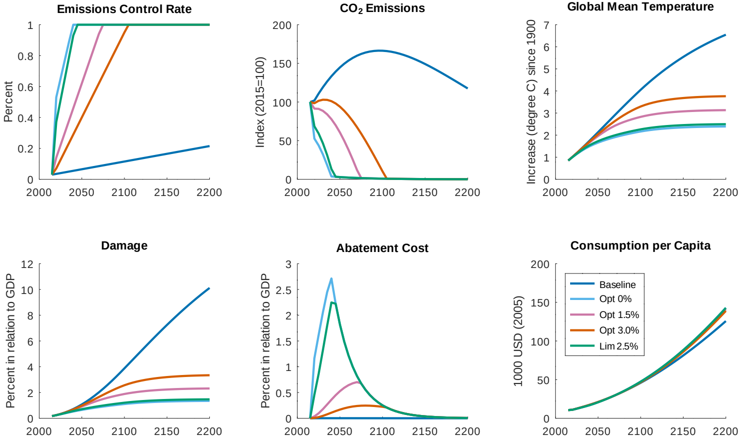
Baseline-Projection

(Nordhaus 2013, 2017, own calculations)



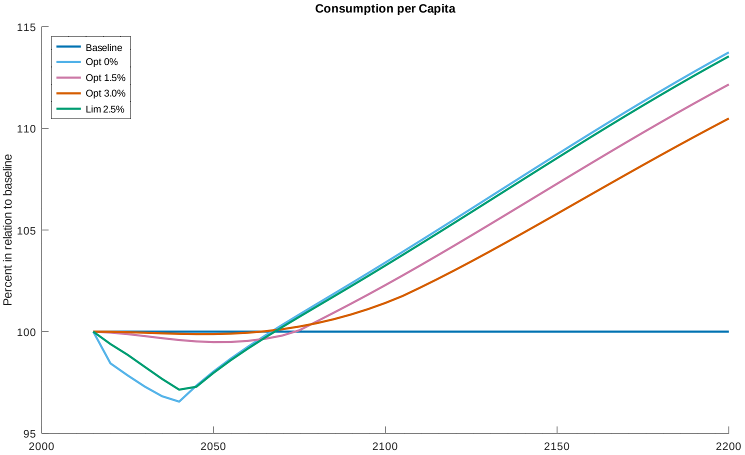
Optimal Scenarios

(Nordhaus 2013, 2017, own calculations)

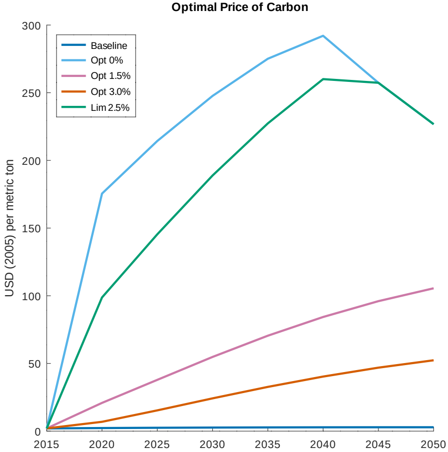


Optimal Climate Policy Implies Less Consumption Today

(Nordhaus 2013, 2017, own calculations)



Efficient Climate Policy



- Economic Optimum: CO_2 price = marginal cost of CO_2-emissions (measured by social welfare W)
- Optimal CO_2 prices 2020:
  - Lim. 2.5: 139 USD/t
  - Opt. 1.5: 34 USD/t
  - Opt. 3.0: 9 USD/t
- National emission trading system:
  - 2023: 30 EUR/t
  - 2026: 55-65 EUR/t

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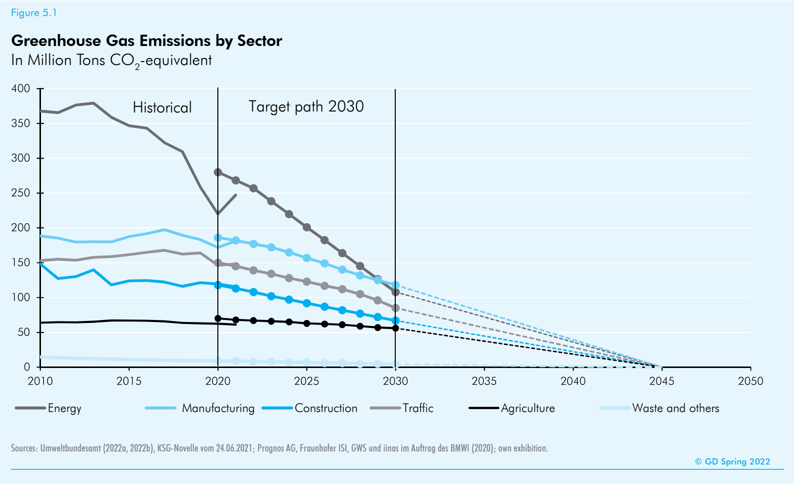
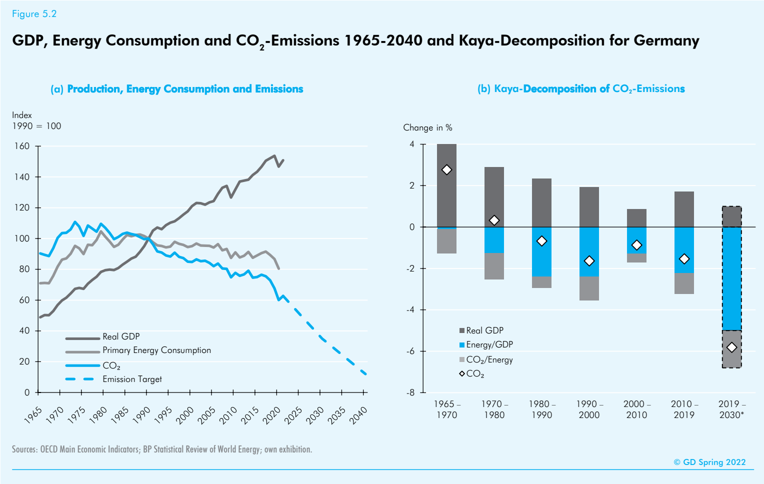
German Greenhouse Gas Emissions



Energy Intensity in Germany



Directed Technical Change as a Response to Natural Resource Scarcity

(Hassler, Krusell, Olovsson 2021)

- Constant-elasticity of substitution (CES) production function

$$Y = \left[(1-\gamma)\left(AK^{\alpha}L^{1-\alpha}\right)^{\frac{\epsilon-1}{\epsilon}} + \gamma\left(A_E E\right)^{\frac{\epsilon-1}{\epsilon}}\right]^{\frac{\epsilon}{\epsilon-1}}$$

  where E denotes Energy use, ϵ elasticity of substitution between capital/labor and energy and γ weight of energy in production

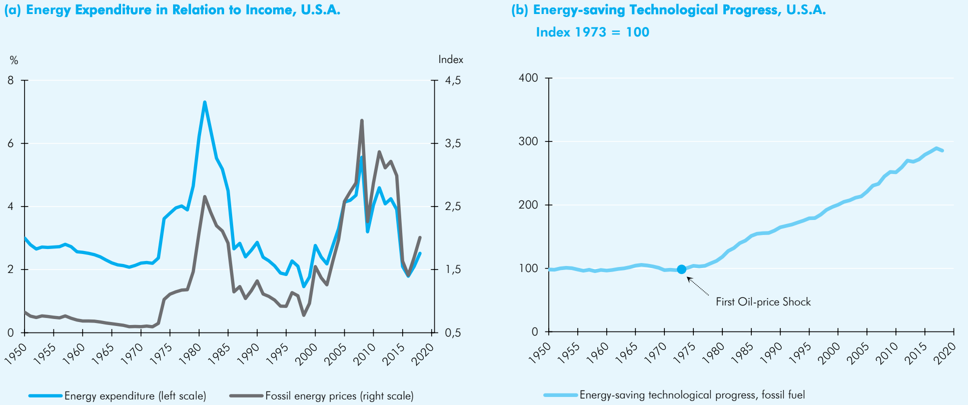
- Estimation of energy-saving technical progress

$$A_E = \frac{Y}{E}\left[\frac{E\text{-Expenditure Share}}{\gamma}\right]^{\frac{\epsilon}{\epsilon-1}}$$

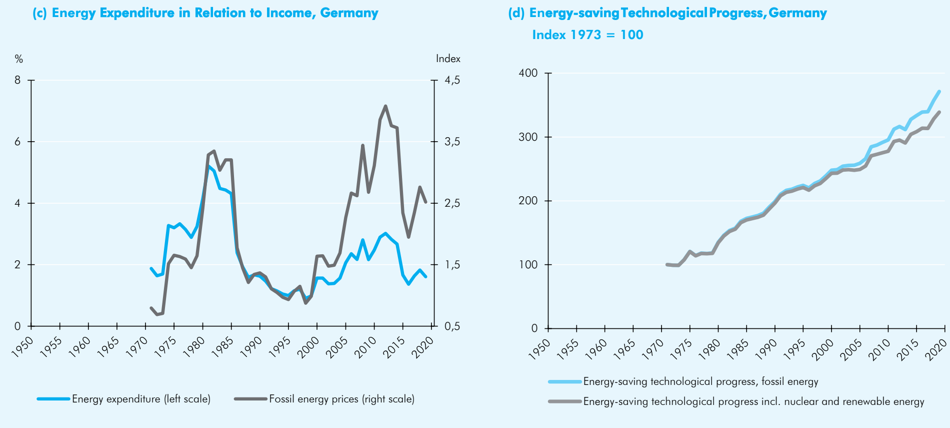
Energy as Production Factor: USA

Figure 5.4

Energy Prices, Energy Expenditure and Energy-saving Technological Progress



Energy as Production Factor: Germany



Sources: International Energy Agency: World Energy Balances; Statistisches Bundesamt; IRENA; Hassler et al. (2021); own calculations.

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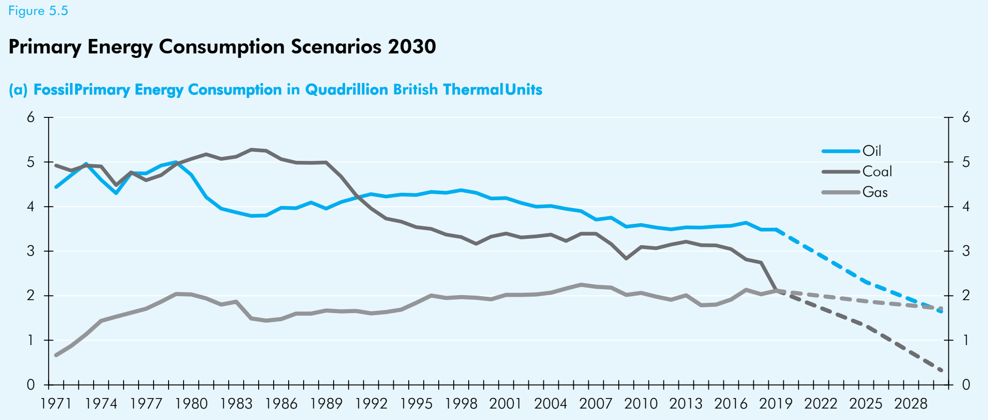
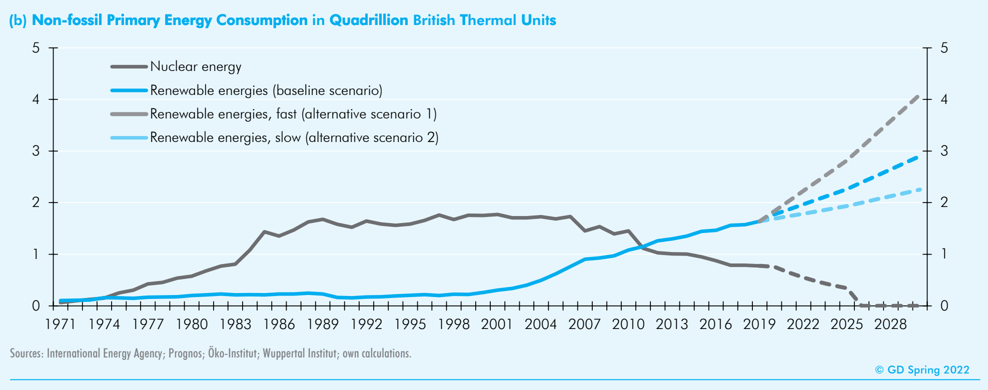
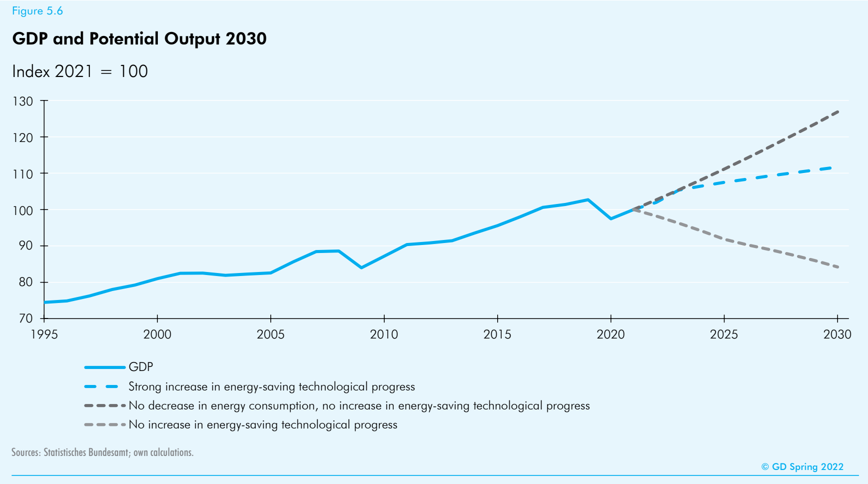
Fossil Energy Consumption in Germany



Non-fossil Energy Consumption: Scenarios for Germany



Potential Output: Scenarios for Germany



Implications

- Rate of energy-saving technological progress
  - 1973–2019: 2.7%
  - Baseline scenario renewable energies: $+5.6\%$
  - Alternative scenario 1 (fast extension): $+4.7\%$
  - Alternative scenario 2 (slow extension): $+6.1\%$
- Potential output 2030
  - Baseline scenario: $+10\%$
  - No increase in energy-saving technological progress: -15%
  - Without decarbonization: $+25\%$

14. Integrated Assessment Modeling

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Summary

- Kaya decomposition: population, production per capita, energy intensity and carbon intensity as drivers of CO_2 emissions.
- Integrated assessment models combine economic growth model and climate change models.
- Energy-saving technical progress important but probably not sufficient to achieve emission reduction targets.

Literature

- Acevedo, S.; Mrkaic, M.; Novta, N.; Pugacheva, E.; Topalova, P. (2018): The Effects of Weather Shocks on Economic Activity: What are the Channels of Impact?, IMF Working Paper 18/144
- Burke, M.; Hsiang, S.M.; Miguel, E. (2015): Global Non-linear Effect of Temperature on Economic Production, Nature, 527, 235-239
- Hassler, J.; Krusell, P.; Olovsson, C. (2021): Directed Technical Change as a Response to Natural Resource Scarcity, Journal of Political Economy 129(11), 3039-3072
- Nordhaus, W.D. (2013): The Climate Casino: Risk, Uncertainty, and Economics for a Warming World, Yale University Press
- Nordhaus, W.D. (2017): Revisiting the Social Cost of Carbon, PNAS 117(7), 1518-1523
- Projektgruppe Gemeinschaftsdiagnose (2022): Von der Pandemie zur Energiekrise, Gemeinschaftsdiagnose Frühjahr 2022, Kiel www.gemeinschaftsdiagnose.de